



Network Design & Management (IT 3010)
3rd Year, 1st Semester

Assignment

**Virtual Network Infrastructure
Implementation using DNS, Mail Server and
TCP Client-Server**

Submitted to
Sri Lanka Institute of Information Technology

IT23557642

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

Student Name: MADALAGAMA U M V S

Student ID: IT23557642

Signature:

A handwritten signature in black ink, appearing to read 'Vimyji', with a long, sweeping horizontal stroke extending to the right.

Date: 01/04/2026

Introduction

This assignment will show how to design and implement a virtual network infrastructure using virtualization. The virtual network infrastructure will be composed of two virtual machines that will be configured to behave like a client-server system.

The implementation includes:

- Static IP configuration based on student registration number (IT23557642)
- DNS server configuration using BIND
- Mail server configuration using Postfix
- Design and implementation of a client-server application using C programming language for TCP communication

The main aim of this assignment is to learn about different network services and communication in a virtual environment.

Network Setup

Two virtual machines were used:

- CentOS Stream 9 – Server
- Fedora 28 – Client

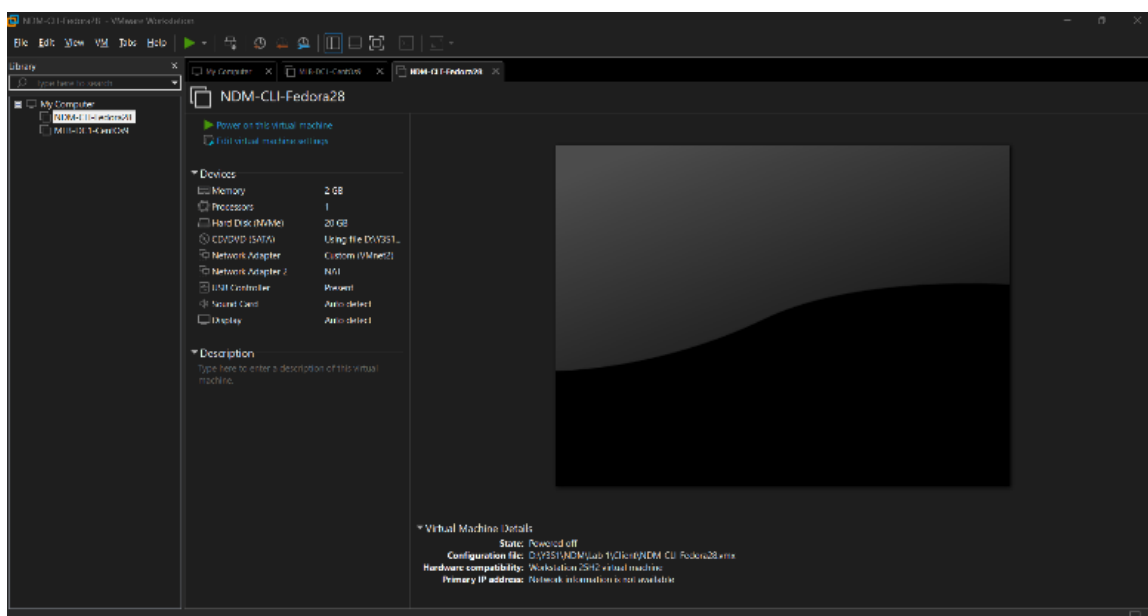
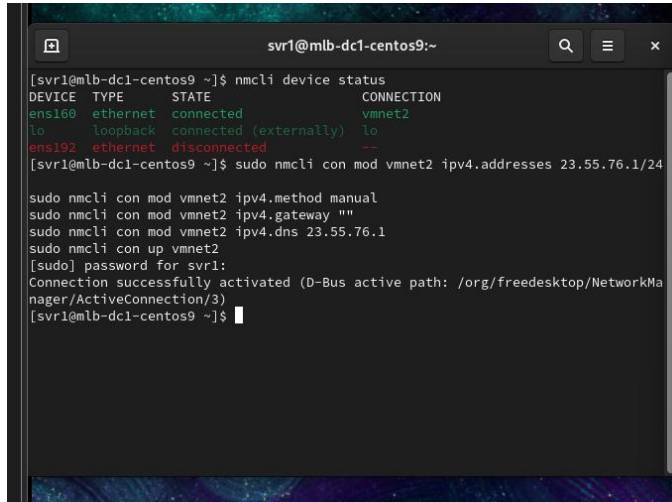


Figure 2.1: VMware virtual machine setup

Network Configuration

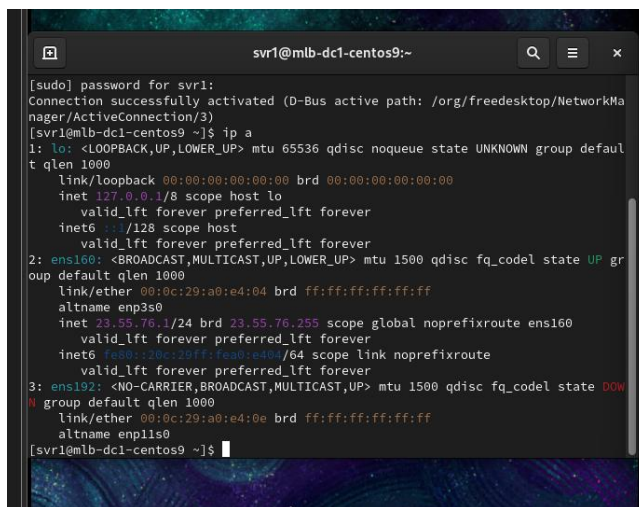
- Network: 23.55.76.0/24
- Server IP: 23.55.76.1
- Client IP: 23.55.76.2



```
svr1@mlb-dc1-centos9:~$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
ens160  ethernet  connected  vmnet2
lo       loopback  connected (externally)  lo
ens192  ethernet  disconnected --
[svr1@mlb-dc1-centos9 ~]$ sudo nmcli con mod vmnet2 ipv4.addresses 23.55.76.1/24

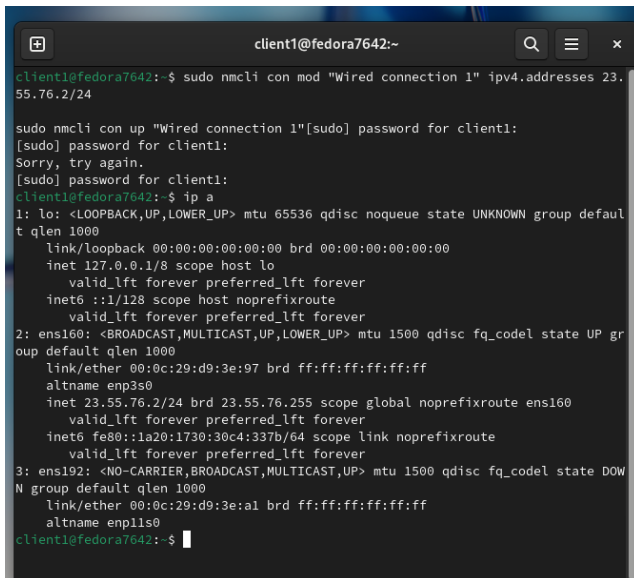
sudo nmcli con mod vmnet2 ipv4.method manual
sudo nmcli con mod vmnet2 ipv4.gateway ""
sudo nmcli con mod vmnet2 ipv4.dns 23.55.76.1
sudo nmcli con up vmnet2
[sudo] password for svr1:
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[svr1@mlb-dc1-centos9 ~]$
```

Figure 2.2: Network adapter configuration



```
[sudo] password for svr1:
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[svr1@mlb-dc1-centos9 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:a0:e4:04 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 23.55.76.1/24 brd 23.55.76.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::a0c:29ff:fe04:4000/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens192: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 00:0c:29:a0:e4:0e brd ff:ff:ff:ff:ff:ff
    altname enp11s0
[svr1@mlb-dc1-centos9 ~]$
```

Figure 2.3: IP configuration of server

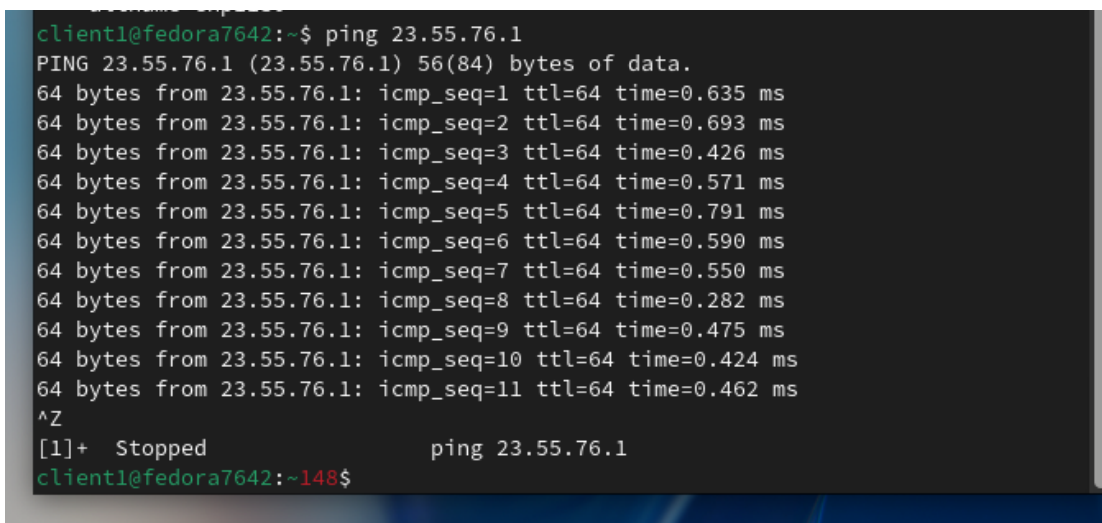


```
client1@fedora7642:~$ sudo nmcli con mod "Wired connection 1" ipv4.addresses 23.55.76.2/24

sudo nmcli con up "Wired connection 1"[sudo] password for client1:
[sudo] password for client1:
Sorry, try again.
[sudo] password for client1:
client1@fedora7642:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d9:3e:97 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 23.55.76.2/24 brd 23.55.76.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::1a20:1730:30c4:337b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens192: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 00:0c:29:d9:3e:a1 brd ff:ff:ff:ff:ff:ff
    altname enp11s0
client1@fedora7642:~$
```

Figure 2.4: IP configuration of client

The VMnet2 network was configured to establish communication between the virtual machines. Static IP addresses were assigned using the `nmcli` command. Network connectivity between the client and server was verified using the `ping` command.



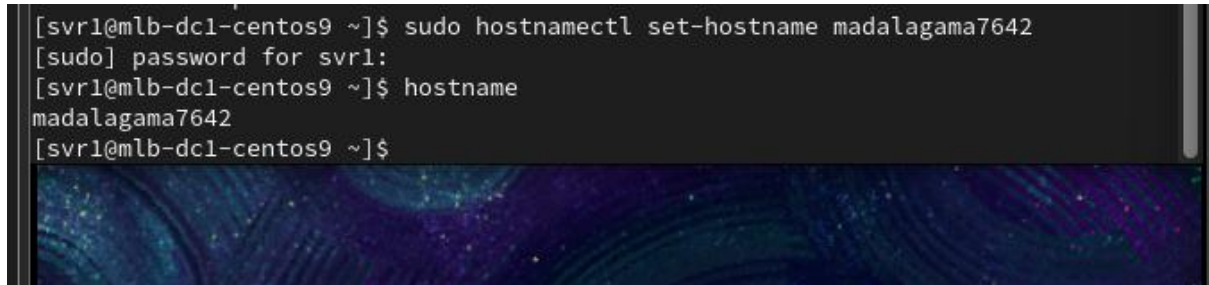
```
client1@fedora7642:~$ ping 23.55.76.1
PING 23.55.76.1 (23.55.76.1) 56(84) bytes of data:
 64 bytes from 23.55.76.1: icmp_seq=1 ttl=64 time=0.635 ms
 64 bytes from 23.55.76.1: icmp_seq=2 ttl=64 time=0.693 ms
 64 bytes from 23.55.76.1: icmp_seq=3 ttl=64 time=0.426 ms
 64 bytes from 23.55.76.1: icmp_seq=4 ttl=64 time=0.571 ms
 64 bytes from 23.55.76.1: icmp_seq=5 ttl=64 time=0.791 ms
 64 bytes from 23.55.76.1: icmp_seq=6 ttl=64 time=0.590 ms
 64 bytes from 23.55.76.1: icmp_seq=7 ttl=64 time=0.550 ms
 64 bytes from 23.55.76.1: icmp_seq=8 ttl=64 time=0.282 ms
 64 bytes from 23.55.76.1: icmp_seq=9 ttl=64 time=0.475 ms
 64 bytes from 23.55.76.1: icmp_seq=10 ttl=64 time=0.424 ms
 64 bytes from 23.55.76.1: icmp_seq=11 ttl=64 time=0.462 ms
^Z
[1]+  Stopped                  ping 23.55.76.1
client1@fedora7642:~$
```

Figure 2.5: Ping test between client and server

Hostnames:

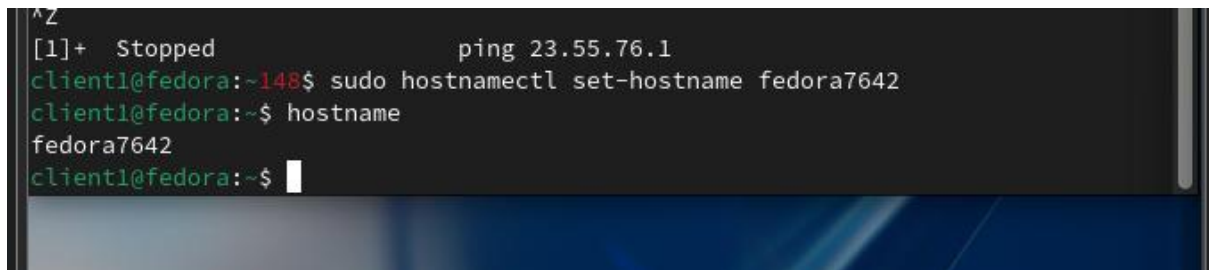
Server: madalagama7642

Client: fedora7642

A terminal window with a dark background and a colorful, abstract pattern at the bottom. The text shows a user 'svr1' on a 'mlb-dcl-centos9' machine running 'sudo hostnamectl set-hostname madalagama7642'. It prompts for a password, then the user runs 'hostname' and the output is 'madalagama7642'.

```
[svr1@mlb-dcl-centos9 ~]$ sudo hostnamectl set-hostname madalagama7642
[sudo] password for svr1:
[svr1@mlb-dcl-centos9 ~]$ hostname
madalagama7642
[svr1@mlb-dcl-centos9 ~]$
```

Figure 2.6: Server hostname

A terminal window with a dark background and a blue abstract pattern at the bottom. The text shows a user 'client1' on a 'fedora' machine. It starts with '[1]+ Stopped ping 23.55.76.1', then the user runs 'sudo hostnamectl set-hostname fedora7642'. After running 'hostname', the output is 'fedora7642'.

```
^Z
[1]+  Stopped                  ping 23.55.76.1
client1@fedora:~148$ sudo hostnamectl set-hostname fedora7642
client1@fedora:~$ hostname
fedora7642
client1@fedora:~$
```

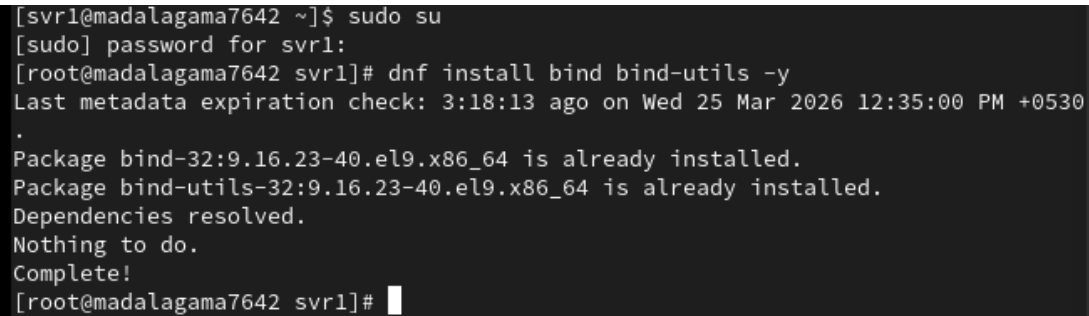
Figure 2.7: Client hostname

DNS Server Configuration

A DNS server was configured on the CentOS server using **BIND**.

Installation:

```
dnf install bind bind-utils -y
```

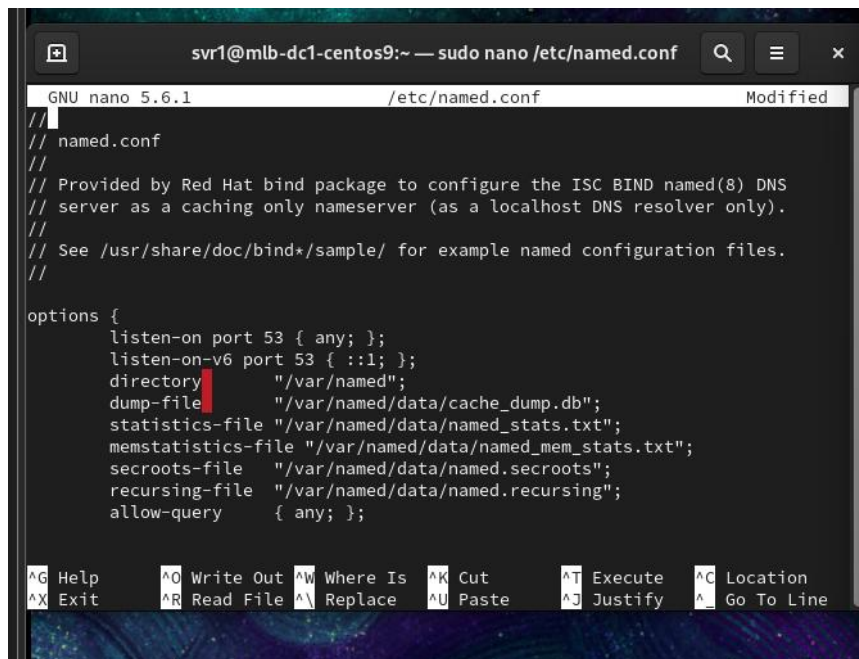
A terminal window showing the installation of BIND. The user runs 'sudo su' to become root, then 'dnf install bind bind-utils -y'. The output shows that both packages are already installed, dependencies are resolved, and the installation is complete.

```
[svr1@madalagama7642 ~]$ sudo su
[sudo] password for svr1:
[root@madalagama7642 svr1]# dnf install bind bind-utils -y
Last metadata expiration check: 3:18:13 ago on Wed 25 Mar 2026 12:35:00 PM +0530
.
Package bind-32:9.16.23-40.el9.x86_64 is already installed.
Package bind-utils-32:9.16.23-40.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@madalagama7642 svr1]#
```

Figure 3.1: BIND installation verification

Configuration:

- Edited /etc/named.conf
- Enabled:
 - listen-on port 53 { any; };
 - allow-query { any; };

A screenshot of the nano text editor editing the /etc/named.conf file. The file contains configuration for the BIND DNS server, including options for listening on port 53 and allowing queries from any host.

```
svr1@mlb-dc1-centos9:~ — sudo nano /etc/named.conf
GNU nano 5.6.1 /etc/named.conf Modified
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { any; };
}
```

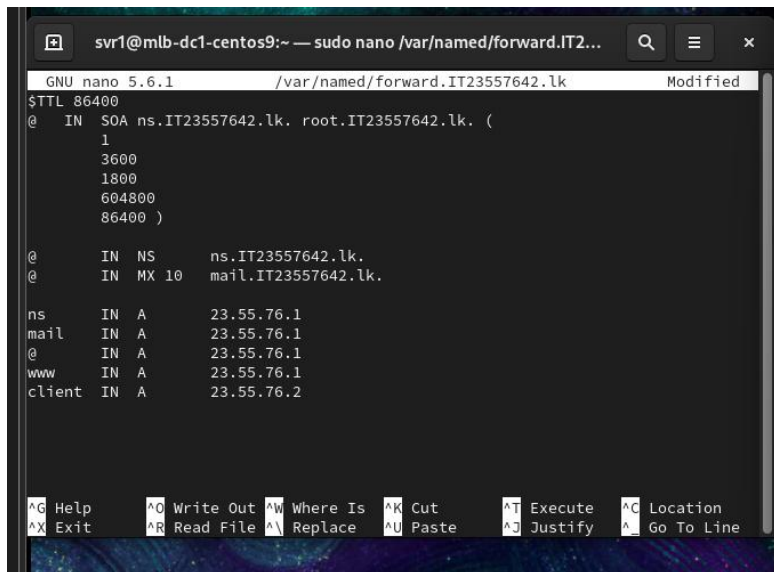
Figure 3.2: named.conf configuration

Zones Created:

- Forward zone – IT23557642.lk
- Reverse zone – 76.55.23.in-addr.arpa

Forward Zone

- The forward lookup zone was configured for the domain IT23557642.lk. A records were used to map hostname (ns, mail, www, client) to IP addresses, and an MX record was added for the mail server.



```
svr1@mlb-dc1-centos9:~ — sudo nano /var/named/forward.IT23557642.lk
GNU nano 5.6.1 /var/named/forward.IT23557642.lk Modified
$TTL 86400
@ IN SOA ns.IT23557642.lk. root.IT23557642.lk. (
    1
    3600
    1800
    604800
    86400 )

@ IN NS ns.IT23557642.lk.
@ IN MX 10 mail.IT23557642.lk.

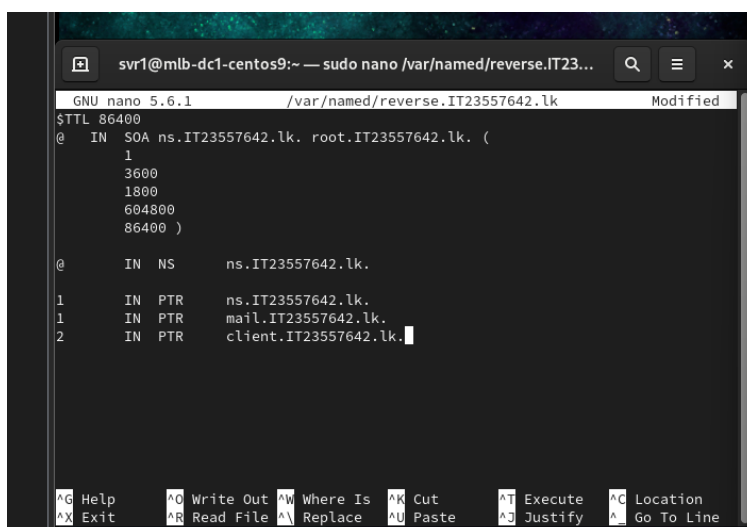
ns IN A 23.55.76.1
mail IN A 23.55.76.1
@ IN A 23.55.76.1
www IN A 23.55.76.1
client IN A 23.55.76.2

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line
```

Figure 3.3: Forward zone configuration

Reverse Zone

- The reverse lookup zone was configured using PTR records to map IP addresses back to their corresponding hostnames.



```
svr1@mlb-dc1-centos9:~ — sudo nano /var/named/reverse.IT23557642.lk
GNU nano 5.6.1 /var/named/reverse.IT23557642.lk Modified
$TTL 86400
@ IN SOA ns.IT23557642.lk. root.IT23557642.lk. (
    1
    3600
    1800
    604800
    86400 )

@ IN NS ns.IT23557642.lk.

1 IN PTR ns.IT23557642.lk.
1 IN PTR mail.IT23557642.lk.
2 IN PTR client.IT23557642.lk.

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line
```

Figure 3.4: Reverse zone configuration

Testing:

- The DNS setup was tested using the **nslookup** command. Both forward and reverse resolutions were successful, confirming that the DNS server is working correctly.

```
[svr1@mlb-dc1-centos9 ~]$ nslookup IT23557642.lk 23.55.76.1
Server:      23.55.76.1
Address:     23.55.76.1#53

Name:   IT23557642.lk
Address: 23.55.76.1

[svr1@mlb-dc1-centos9 ~]$
```

Figure 3.5: Forward DNS lookup for IT23557642

```
[svr1@mlb-dc1-centos9 ~]$ nslookup mail.IT23557642.lk 23.55.76.1
Server:      23.55.76.1
Address:     23.55.76.1#53

Name:   mail.IT23557642.lk
Address: 23.55.76.1

[svr1@mlb-dc1-centos9 ~]$
```

Figure 3.6: Forward DNS lookup for mail server

```
[svr1@mlb-dc1-centos9 ~]$ nslookup 23.55.76.1 23.55.76.1
1.76.55.23.in-addr.arpa name = mail.IT23557642.lk.
1.76.55.23.in-addr.arpa name = ns.IT23557642.lk.

[svr1@mlb-dc1-centos9 ~]$
```

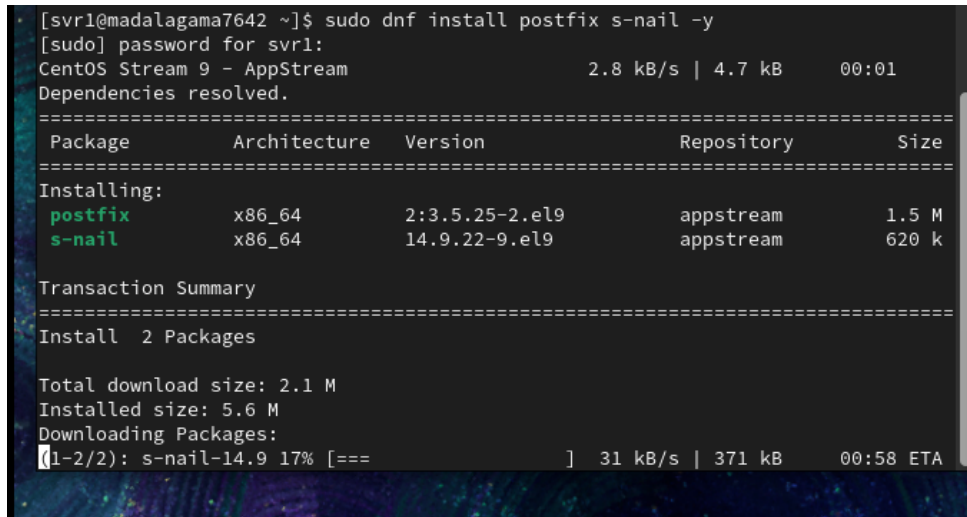
Figure 3.7: Reverse DNS lookup result

Mail Server Configuration

A mail server was implemented using **Postfix**.

Installation:

```
dnf install postfix s-nail -y
```



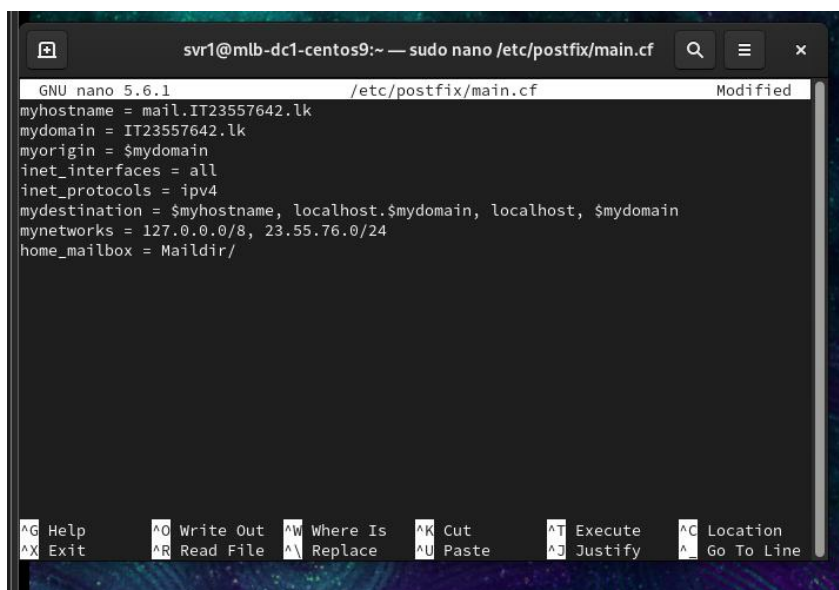
```
[svr1@madalagama7642 ~]$ sudo dnf install postfix s-nail -y
[sudo] password for svr1:
CentOS Stream 9 - AppStream                2.8 kB/s | 4.7 kB    00:01
Dependencies resolved.
=====
Package            Architecture Version           Repository        Size
=====
Installing:
postfix            x86_64         2:3.5.25-2.el9   appstream         1.5 M
s-nail             x86_64         14.9.22-9.el9    appstream         620 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 2.1 M
Installed size: 5.6 M
Downloading Packages:
[1-2/2]: s-nail-14.9 17% [===          ] 31 kB/s | 371 kB    00:58 ETA
```

Figure 4.1: Postfix installation

Configuration:

- Edited `/etc/postfix/main.cf`
 - `myhostname = mail.IT23557642.lk`
 - `mydomain = IT23557642.lk`
 - `inet_interfaces = all`
 - `mydestination configured`
 - `home_mailbox = Maildir/`



```
svr1@mlb-dc1-centos9:~ — sudo nano /etc/postfix/main.cf
GNU nano 5.6.1 /etc/postfix/main.cf Modified
myhostname = mail.IT23557642.lk
mydomain = IT23557642.lk
myorigin = $mydomain
inet_interfaces = all
inet_protocols = ipv4
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mynetworks = 127.0.0.0/8, 23.55.76.0/24
home_mailbox = Maildir/

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Figure 4.2: Postfix main.cf configuration

User Setup:

- Created users: user1, user2
- Created Maildir directories

```
[svr1@madalagama7642 ~]$ sudo useradd user1
[svr1@madalagama7642 ~]$ sudo passwd user1
Changing password for user user1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
[svr1@madalagama7642 ~]$ sudo useradd user1
useradd: user 'user1' already exists
[svr1@madalagama7642 ~]$ sudo passwd user1
Changing password for user user1.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/sys
tematic
Retype new password:
passwd: all authentication tokens updated successfully.
[svr1@madalagama7642 ~]$
```

Figure 4.3: Creating user1 and setting password

```
[svr1@madalagama7642 ~]$ sudo useradd user2
[svr1@madalagama7642 ~]$ sudo passwd user2
Changing password for user user2.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/sys
tematic
Retype new password:
passwd: all authentication tokens updated successfully.
[svr1@madalagama7642 ~]$
```

Figure 4.4: Creating user2 and setting password

```
[svr1@madalagama7642 ~]$ su - user1
Password:
[user1@madalagama7642 ~]$ mkdir -p ~/Maildir/{new,cur,tmp}
[user1@madalagama7642 ~]$ exit
logout
[svr1@madalagama7642 ~]$
```

Figure 4.5: Creating Maildir for user1

```
[svr1@madalagama7642 ~]$ su - user2
Password:
[user2@madalagama7642 ~]$ mkdir -p ~/Maildir/{new,cur,tmp}
[user2@madalagama7642 ~]$ exit
logout
[svr1@madalagama7642 ~]$
```

Figure 4.6: Creating Maildir for user2

Testing:

- Email communication was tested by sending an email from **user1** to **user2** using the mail command. The delivery was verified by checking the Maildir directory, where the received email was successfully stored and displayed.

```
[user2@madalagama7642 ~]$ ls -l ~/Maildir/new
total 4
-rw-----. 1 user2 user2 426 Mar 24 23:42 1774375947.Vfd00I21d32f8M929125.madal
agama7642
[user2@madalagama7642 ~]$ cat ~/Maildir/new/*
Return-Path: <user1@IT23557642.lk>
X-Original-To: user2
Delivered-To: user2@IT23557642.lk
Received: by mail.IT23557642.lk (Postfix, from userid 1001)
        id DB2AB107C687; Tue, 24 Mar 2026 23:42:27 +0530 (+0530)
Date: Tue, 24 Mar 2026 23:42:27 +0530
To: user2@IT23557642.lk
Subject: Local Test
User-Agent: s-nail v14.9.22
Message-Id: <20260324181227.DB2AB107C687@mail.IT23557642.lk>
From: user1@IT23557642.lk

Local delivery test
[user2@madalagama7642 ~]$
```

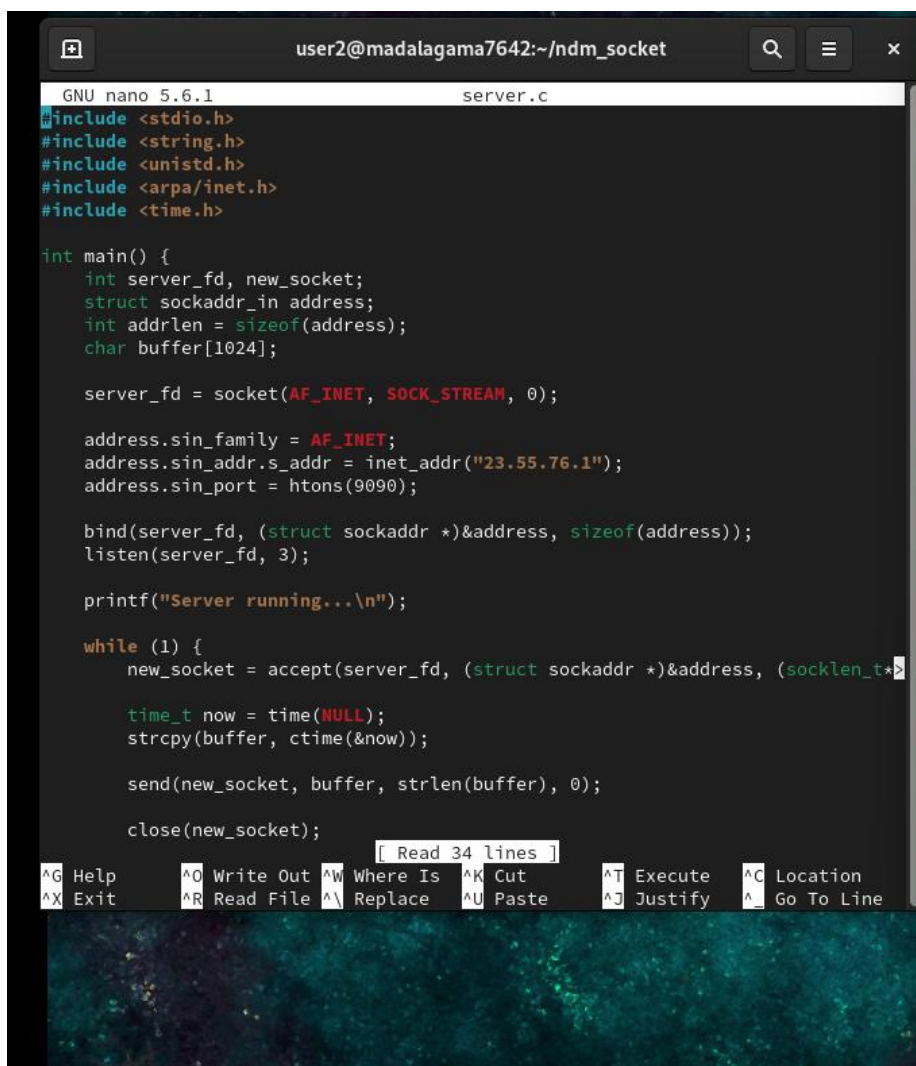
Figure 4.7: Email received and verified using Maildir

Client-Server Application

A TCP client-server application was implemented in C to demonstrate communication between two systems in a network environment.

Server:

- The server program was configured to listen on port 9090 for incoming client connections. Once a connection is established, the server retrieves the current system date and time and sends it to the connected client. The server operates in an iterative manner, handling one client request at a time.



```
GNU nano 5.6.1 server.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>

int main() {
    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);
    char buffer[1024];

    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    address.sin_family = AF_INET;
    address.sin_addr.s_addr = inet_addr("23.55.76.1");
    address.sin_port = htons(9090);

    bind(server_fd, (struct sockaddr *)&address, sizeof(address));
    listen(server_fd, 3);

    printf("Server running...\n");

    while (1) {
        new_socket = accept(server_fd, (struct sockaddr *)&address, (socklen_t*)&);

        time_t now = time(NULL);
        strcpy(buffer, ctime(&now));

        send(new_socket, buffer, strlen(buffer), 0);

        close(new_socket);
    }
}
```

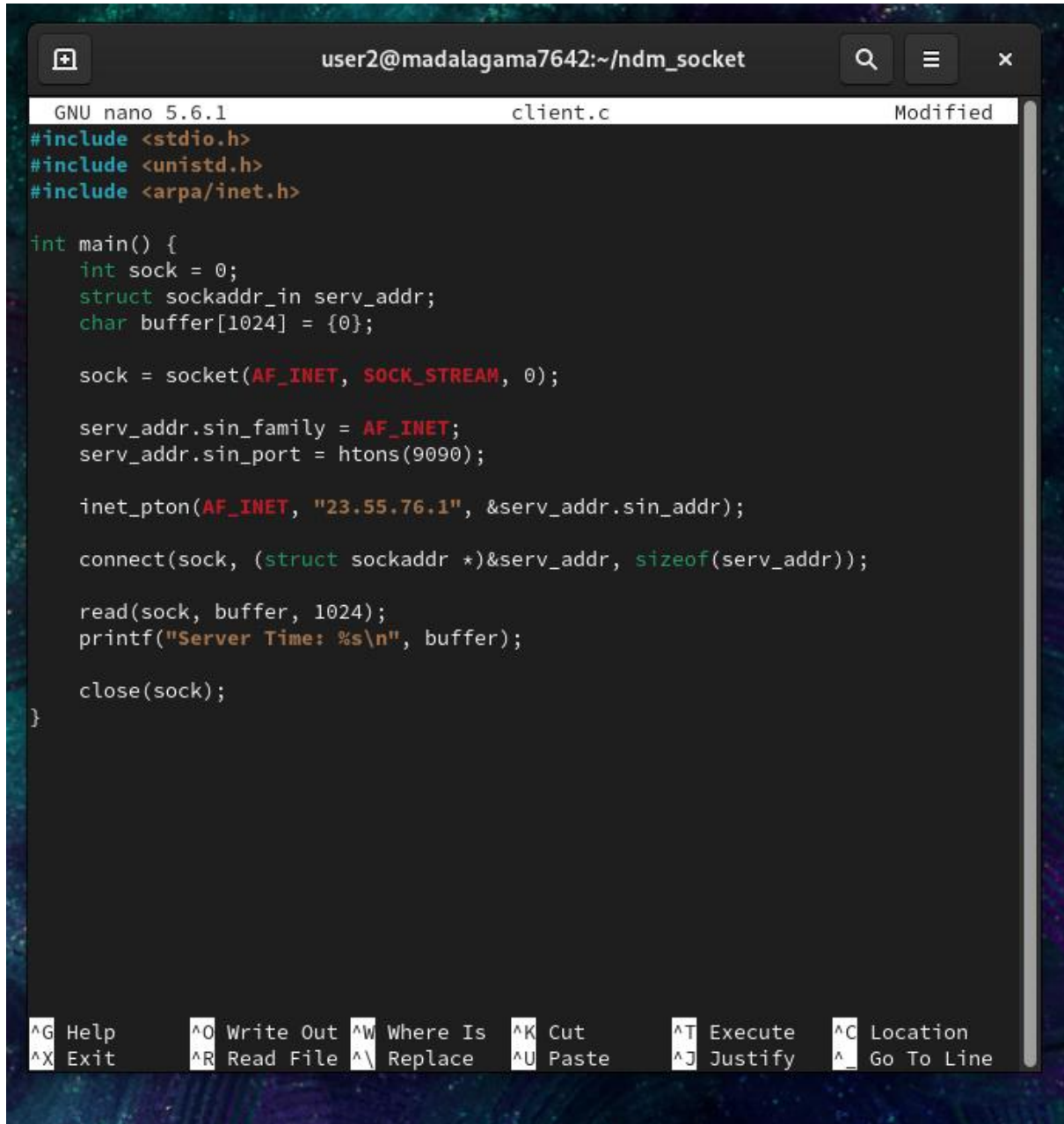
[Read 34 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Figure 5.1: Server program implementation in C

Client:

- The client program connects to the server using the server's IP address and port number. After establishing the connection, it receives the date and time sent by the server and displays the output on the terminal.



```
GNU nano 5.6.1 client.c Modified
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
    int sock = 0;
    struct sockaddr_in serv_addr;
    char buffer[1024] = {0};

    sock = socket(AF_INET, SOCK_STREAM, 0);

    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons(9090);

    inet_pton(AF_INET, "23.55.76.1", &serv_addr.sin_addr);

    connect(sock, (struct sockaddr *)&serv_addr, sizeof(serv_addr));

    read(sock, buffer, 1024);
    printf("Server Time: %s\n", buffer);

    close(sock);
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Figure 5.2: Client program implementation in C

Testing:

- The server application was executed on the CentOS virtual machine, while the client was executed on the Fedora (or CentOS) virtual machine. The client successfully connected to the server and displayed the current date and time, confirming that the communication between the client and server was functioning correctly.

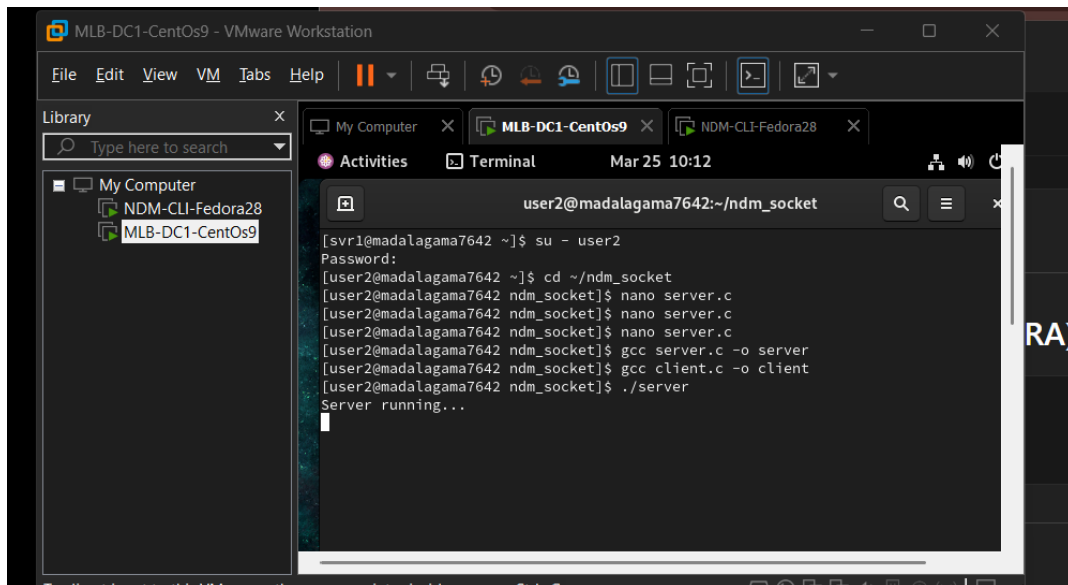


Figure 5.3: Server running on CentOS

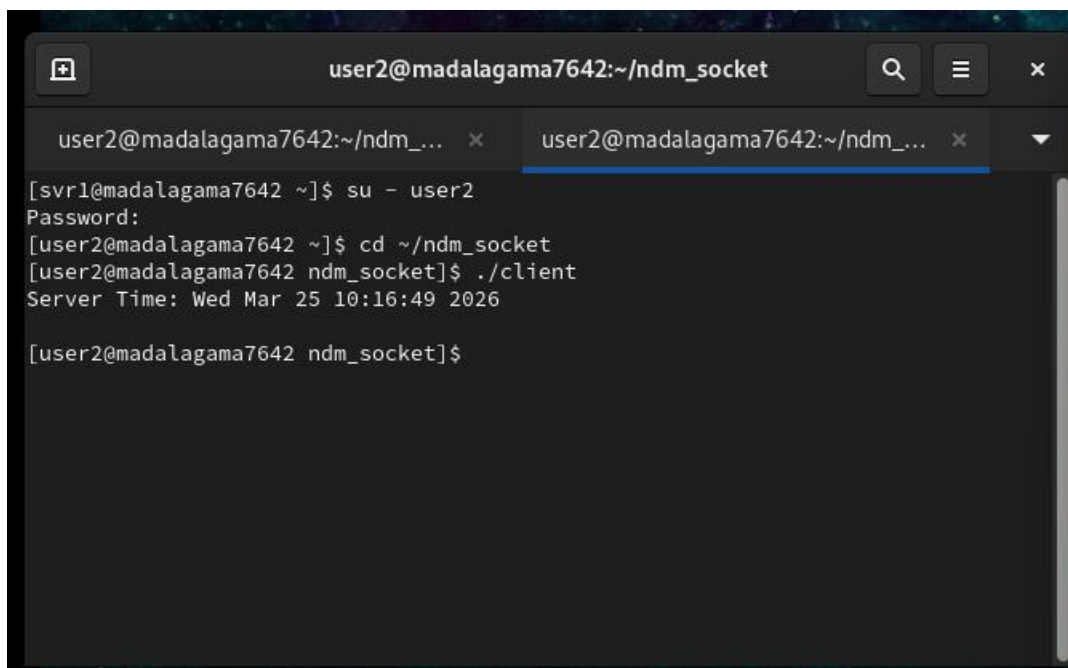


Figure 5.4: Client receiving date and time from server

Conclusion

This assignment was successfully used to demonstrate the setup of a virtual network infrastructure, which included a DNS server, a mail server, and a TCP client-server application. This practical was useful in that I was able to learn how to configure network services, solve problems, and understand client-server communication.